

Hardy-Weinberg Practice

The Hardy-Weinberg formulas allow scientists to determine whether evolution has occurred. Any changes in the gene frequencies in the population over time can be detected. The law essentially states that if no evolution is occurring, then an equilibrium of allele frequencies will remain in effect in each succeeding generation of sexually reproducing individuals. In order for equilibrium to remain in effect (i.e. that no evolution is occurring) then the following five conditions must be met:

1. No mutations must occur so that new alleles do not enter the population.
2. No gene flow can occur (i.e. no migration of individuals into, or out of, the population).
3. Random mating must occur (i.e. individuals must pair by chance)
4. The population must be large so that no genetic drift (random chance) can cause the allele frequencies to change.
5. No selection can occur so that certain alleles are not selected for, or against.

Obviously, the Hardy-Weinberg equilibrium cannot exist in real life. Some or all of these types of forces all act on living populations at various times and evolution at some level occurs in all living organisms. The Hardy-Weinberg formulas allow us to detect some allele frequencies that change from generation to generation, thus allowing a simplified method of determining that evolution is occurring. There are two formulas that must be memorized:

$$p^2 + 2pq + q^2 = 1 \text{ and } p + q = 1$$

p = frequency of the dominant allele in the gene pool

q = frequency of the recessive allele in the gene pool

p² = percentage of homozygous dominant individuals

q² = percentage of homozygous recessive individuals

2pq = percentage of heterozygous individuals

} Make up the population

PROBLEM #1.

You have sampled a population in which you know that the percentage of the homozygous recessive genotype (aa) is 36%. Using that 36%, calculate the following:

- A. The frequency of the "aa" genotype.
- B. The frequency of the "a" allele.
- C. The frequency of the "A" allele.
- D. The frequencies of the genotypes "AA" and "Aa."
- E. The frequencies of the two possible phenotypes if "A" is completely dominant over "a."

